B. Prove it yourself using Ordered Resolution, where symbols earlier in alphabetical order are larger.

Resources:

I also took a class on propositional logic years ago, which I tried to draw from.

http://intrologic.stanford.edu/chapters/chapter_06.html

https://discrete.openmathbooks.org/dmoi3/sec_propositional.html

Logical Equivalence

You might have noticed in Example 3.1.1 that the final column in the truth table for $\neg P \vee Q$ is identical to the final column in the truth table for $P \to Q$:

P	Q	$P \to Q$	$\neg P \vee Q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

This says that no matter what P and Q are, the statements $\neg P \vee Q$ and $P \to Q$ either both true or both false. We therefore say these statements are ***logically equivalent***.

---

1. Last night, there was a party. Ann and Bill and Carl and Dave were invited. Whenever, there is a party, Carl always attends when Ann and Bill attend. But Carl will not attend if Dave attends. We know that Ann attended the party. Prove that either Bill or Dave did not attend the party.

Atoms:	Premises/Goal:
Party = party happened Ann = ann attended Bill = bill attended Car = carl attended Dave = dave attended	Party (party ∧ Ann ∧ Bill) -> Carl Dave -> ¬carl Ann ∴¬bill ∨¬dave

Ordered-resolution proof

Ann > Bill > Carl > Dave > Party

1. Party (by premise)
2. Ann (by premise)
3. (party ∧ Ann ∧ Bill) -> Carl (by premise)
4. Dave -> ¬carl (by premise)
5. ¬party V ¬Ann V ¬Bill V Carl (by logical equivalence, line 3 used)
6. ¬Dave V ¬Carl (by logical equivalence, line 4 used)
7. Bill (by assumption, negated goal, trying to get contradiction)
8. Dave (by assumption, negated goal, trying to get contradiction)
9. ¬Party V ¬Bill V Carl (By resolution 2, 5 on ann)
10. ¬Party V Carl (By resolution 7, 9 on bill)
11. ¬Carl (By resolution 6, 8 on dave)
12. ¬Party (By resolution 10, 11 on carl)
13. $F (By resolution 1, 12)

Since the premises cannot all be true if both Bill and Dave attended, they cannot both have attended.

∴¬bill v¬dave

---

2. There was another party. Ann and Bill and Carl and Dave

were invited. Not all of them attended. Whenever there is a party,

Bill attends when Ann attends. Also, Carl attends when Bill attends.

Also, Dave attends when Carl attends. Also, Ann attends when Dave attends.

Prove that nobody came to the party.

Atoms:	Premises/Goal:
Party = party happened Ann = ann attended Bill = bill attended Carl = carl attended Dave = dave attended	Party ¬Ann V ¬Bill V ¬Carl V ¬Dave Ann -> Bill Bill -> Carl Carl -> Dave Dave-> Ann

	∴¬Ann ∧ ¬Bill ∧¬Carl ∧¬Dave

Ordered-resolution proof

Ann > Bill > Carl > Dave > Party

Assuming Ann attended

1. ¬Ann V ¬Bill V ¬Carl V ¬Dave (by premise)
2. Ann -> Bill (by premise)
3. Bill -> Carl (by premise)
4. Carl -> Dave (by premise)
5. Dave-> Ann (by premise)
6. ¬Ann V Bill (logical equivalence 2)
7. ¬Bill V Carl (logical equivalence 3)
8. ¬Carl V Dave (logical equivalence 4)
9. ¬Dave V Ann (logical equivalence 5)
10. Ann (By assumption)
11. Bill (By resolve 6, 10)
12. Carl (By resolution 7,11)
13. Dave (By resolution 8,12)
14. ¬Bill V ¬Carl V ¬Dave (By resolution 1, 10 on ann)
15. ¬Carl V ¬Dave (By resolution 11, 14 on bill)
16. ¬Dave (By resolution 12, 15 on carl)
17. $F (by resolution of 13 and 16 on Dave)

Assuming Bill attended

1. ¬Ann V ¬Bill V ¬Carl V ¬Dave (by premise)
2. ¬Ann V Bill (logical equivalence 2)
3. ¬Bill V Carl (logical equivalence 3)
4. ¬Carl V Dave (logical equivalence 4)
5. ¬Dave V Ann (logical equivalence 5)
6. Bill (By assumption)
7. Carl (By resolution 3, 6)
8. Dave (By resolution 4,7)
9. Ann (By resolution 5,8)
10. ¬Ann V ¬Carl V ¬Dave (By resolution 1,6)

11. ¬Ann V ¬Dave (By resolution 7, 10)
12. ¬Ann (By resolution 8, 11)
13. $F (By resolution 9, 12)

Assuming Carl attended

1. ¬Ann V ¬Bill V ¬Carl V ¬Dave (by premise)
2. ¬Ann V Bill (logical equivalence 2)
3. ¬Bill V Carl (logical equivalence 3)
4. ¬Carl V Dave (logical equivalence 4)
5. ¬Dave V Ann (logical equivalence 5)
6. Carl (By assumption)
7. Dave (by resolution 4, 6)
8. Ann (resolution 5,7)
9. Bill (resolution 2, 8)
10. ¬Ann V ¬Bill V ¬Dave (resolution 1,6)
11. ¬Ann V ¬Bill (resolve 10 and 7 on Dave)
12. ¬Ann (resolve 11 and 9 on Bill)
13. $F (resolve 12 and 8 on Ann)

Assuming Dave attended

1. ¬Ann V ¬Bill V ¬Carl V ¬Dave (by premise)
2. ¬Ann V Bill (logical equivalence 2)
3. ¬Bill V Carl (logical equivalence 3)
4. ¬Carl V Dave (logical equivalence 4)
5. ¬Dave V Ann (logical equivalence 5)
6. Dave (By assumption)
7. Ann (resolution 5,6)
8. Bill (resolution 2,7)
9. Carl (resolution 3, 8)
10. ¬Ann V ¬Bill V ¬Carl (resolution 1,6)
11. ¬Bill V ¬Carl (resolution 7,10)
12. ¬Carl (resolution 8,11)
13. $F (By resolution 9, 12)

Each case assumes one person attended and all end in a contradiction so

∴¬Ann ∧ ¬Bill ∧¬Carl ∧¬Dave

---

3. There was another party. Ann and Bill and Carl and Dave

were invited. There were at least two people at the party.

If Ann attended then everybody else attended. And Bill always

attends when Carl attends. Prove that Bill came to the party.

Atoms:	Premises/Goal:
Party = party happened Ann = ann attended Bill = bill attended Carl = carl attended Dave = dave attended	Party At least two attended: (Ann ∧ Bill) ∨ (Ann ∧ Carl) ∨ (Ann ∧ Dave) ∨ (Bill ∧ Carl) ∨ (Bill ∧ Dave) ∨ (Carl ∧ Dave) Ann -> Bill Ann -> Carl Ann -> Dave Carl -> Bill ∴ Bill

Ordered-resolution proof

Ann > Bill > Carl > Dave > Party

Assuming Bill did not attend

1. Ann V Bill V Carl (at least two attended)
2. ¬Ann V Bill (Ann -> Bill)
3. ¬Carl V Bill (Carl -> Bill)
4. ¬Bill (negated goal)
5. ¬Ann (resolve 2 and 4 on Bill)
6. ¬Carl (resolve 3 and 4 on Bill)
7. Ann V Carl (resolve 1 and 4 on Bill)
8. Carl (resolve 7 and 5 on Ann)
9. $F (resolve 8 and 6 on Carl)

Since assuming ¬Bill leads to a contradiction, Bill must have attended the party.

4. There was another party. This time only Ann and Bill and Carl were invited. Either Ann or Bill attended, but not both. Either Bill or Carl attended, but not both. Prove if either Ann or Carl came to the party, then both Ann and Carl came to the party.

Atoms:	Premises/Goal:
Party = party happened Ann = ann attended Bill = bill attended Carl = carl attended Dave = dave attended	Party (annVbill)Λ(¬annV¬bill) (billVcarl)Λ(¬billV¬carl) ∴ (ann V carl) -> (ann Λ carl)

Proof:

Ann > Bill > Carl > Dave > Party

Assuming ann attended

1. Ann ∨ Bill (by premise)
2. ¬Ann ∨ ¬Bill (by premise)
3. Bill ∨ Carl (by premise)
4. ¬Bill ∨ ¬Carl (by premise)
5. Ann ∨ Carl (by premise)
6. ¬Ann ∨ ¬Carl (by premise)
7. Ann (by assumption, negated goal)
8. ¬Bill (by resolution of lines 2 and 7 on Ann)
9. Carl (by resolution of lines 3 and 8 on Bill)
10. ¬Carl (by resolution of lines 6 and 7 on Ann)
11. \F$ (by resolution of lines 9 and 10 on Carl)

Assuming not ann

1. Ann ∨ Bill (by premise)
2. ¬Ann ∨ ¬Bill (by premise)
3. Bill ∨ Carl (by premise)
4. ¬Bill ∨ ¬Carl (by premise)
5. Ann ∨ Carl (by premise)
6. ¬Ann ∨ ¬Carl (by premise)
7. ¬Ann (by assumption)
8. Bill (by resolution of lines 1 and 7 on Ann)
9. ¬Carl by resolution of lines 4 and 8 on Bill)
10. Carl (by resolution of lines 5 and 7 on Ann)
11. \F$ (by resolution of lines 9 and 10 on Carl)

Assuming Ann attended and did not attend the party leads to a contradiction, so the negated goal is false and the original goal is proved.